

AMD RYZEN™ IN 2022

CONSUMER NOTEBOOK PROCESSOR LINEUP

H-SERIES

FOR GAMERS & CREATORS

AMD Model	Cores/ Threads	Max Boost (Base)	L2+L3 Cache	GPU Cores (Max Boost)	Node	TDP
RYZEN™ 9 6980HX	8/16	5.0 (3.3)	20MB	12 (2.4GHz)	6nm	45W+
RYZEN™ 9 6980HS	8/16	5.0 (3.3)	20MB	12 (2.4GHz)	6nm	35W
RYZEN™ 9 6900HX	8/16	4.9 (3.3)	20MB	12 (2.4GHz)	6nm	45W+
RYZEN™ 9 6900HS	8/16	4.9 (3.3)	20MB	12 (2.4GHz)	6nm	35W
RYZEN™ 7 6800H	8/16	4.7 (3.2)	20MB	12 (2.2GHz)	6nm	45W
RYZEN™ 7 6800HS	8/16	4.7 (3.2)	20MB	12 (2.2GHz)	6nm	35W
RYZEN™ 5 6600H	6/12	4.5 (3.3)	19MB	6 (1.9GHz)	6nm	45W
RYZEN™ 5 6600HS	6/12	4.5 (3.3)	19MB	6 (1.9GHz)	6nm	35W

U-SERIES

THIN & POWERFUL MOBILITY

AMD Model	Cores/ Threads	Max Boost (Base)	L2+L3 Cache	GPU Cores (Max Boost)	Node	TDP
RYZEN™ 7 6800U	8/16	4.7 (2.7)	20MB	12 (2.2GHz)	6nm	15-28W
RYZEN™ 5 6600U	6/12	4.5 (2.9)	19MB	6 (1.9GHz)	6nm	15-28W
RYZEN™ 7 5825U	8/16	4.5 (2.0)	20MB	8 (1.8GHz)	7nm	15W
RYZEN™ 5 5625U	6/12	4.3 (2.3)	19MB	7 (1.6GHz)	7nm	15W
RYZEN™ 3 5425U	4/8	4.1 (2.2)	10MB	6 (1.5GHz)	7nm	15W

AMD has announced a lineup of eight Ryzen H-class CPUs with core counts ranging from 6 to 8. As for the U-series, when comparing the Ryzen 7 6800U to a Ryzen 7 5800U, the results look even more beguiling, with upto 1.7x improvement in Puget Systems Adobe Premiere Pro test and a 2.3x gain in Blender’s GPU Benchmark. The RDNA 2 graphics cores in the Ryzen 6000 will also provide considerable gaming improvements, according to AMD, with a Ryzen 7 6800U offering almost twice the 1080p gaming performance of the Ryzen 7 5800U. In fact, AMD claims that the Ryzen 6000’s new RDNA 2 cores provide a 1.2x to 3x performance boost over Intel’s top Core i7-1165G7 Iris Xe graphics, as well as higher performance than Nvidia’s GeForce MX450 GPU.

Intel

Intel’s long-awaited mobile Alder Lake CPUs have set the stage for some spectacular battle in 2022. Intel has already promised a performance improvement of more than 28 percent over Tiger Lake, and there’s always a slew of Intel-powered laptops to choose from.

Alder-Lake 12th Gen CPUs

At CES 2022, Intel unveiled its new

Alder Lake 12th-generation Core H-series mobile processors, paving the way for a new generation of gaming notebooks this year. With its mix of performance and efficiency cores, Intel described the Alder Lake architecture earlier this year. The mobile H-series range has eight processors with 8 and 14 total cores, grouped into two sets of 4 and 6 performance cores. The Intel Core i9-12900 HK will be the flagship processor, with a Core i5-12450H at the bottom. The new “P-series” processor class, which will handle what the chip manufacturer calls “performance thin-and-light laptops,” comes along as an unexpected surprise. At CES 2022,

Intel announced thirteen mainstream 12th-generation Core processors, ranging from a Core i9 to a Celeron processor, as well as nine “T”-series CPUs geared for low-power 35W systems. Intel also unveiled the 12th-generation “KS” CPU, which would have a clock speed of up to 5.5 GHz.

Arc GPUs

As the market strives for more GPUs, the Intel Arc GPUs were also welcomed with open hands. However, there was nothing very exciting in terms of presented performance estimates, which might not be pleasing to users looking forward to the Arc series as a heavy graphic alternative.

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